TT-P-37C
November 27, 1968
SUPERSEDING
Int. Fed. Spec. TT-P-0037B(HUD-HAA)
August 16, 1967

FEDERAL SPECIFICATION

PAINT, ALKYD RESIN; EXTERIOR TRIM, DEEP COLORS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers alkyd resin ready-mixed paint in deep colors for use on primed wood trim, doors, shutters, and primed metal trim or doors.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

TT-P-143 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of.
TT-P-442 - Pigment, Titanium Dioxide (For Protective Coatings).
TT-R-266 - Resin, Alkyd; Solutions.
TT-T-291 - Thinner; Paint, Volatile Mineral Spirits (Petroleum-Spirits).

Federal Standards:

Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer and Related Materials; Methods of inspection, Sampling and Testing.
Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

3. REQUIREMENTS

3.1 Material. The paint as received shall consist of pigment or pigments and vehicle specified, so combined as to produce a paint meeting all the requirements of this specification.

3.2 Color. The color of the paint when tested as in 4.3.12 shall match the standard color chip in Fed. Std. No. 595.

3.3 Composition.

3.3.1 Pigments. The pigments listed in Table II, or any combination thereof, shall make up the basic pigmentation for the color specified. Prime pigments shall be of a good commercial quality. Tinting pigments may be used when necessary to match the color chips provided these additional pigments have good color permanence. The titanium dioxide shall be rutile, chalk-resisting type conforming to type III or Type IV of TT-P-442. Extender pigments may be used at the discretion of the manufacturer provided the paint conforms to all requirements.

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3.3.2 Vehicle. The vehicle shall be alkyd resin conforming to type I, either Class A, B, or C of TT-R-266, together with necessary thinners and driers. Small amounts of antiskinning agents, wetting agents, suspension agents, and anti absorption agents may be added at the discretion of the manufacturer. Solvents or solvent system used shall comply Rule 66 1/.

3.4 Qualitative requirements.

3.4.1 Condition in container. A freshly opened full container of the paint tested as in 4.3.3 shall show no skinning, thickening, livering, gelatinous masses, ingredient separation, and hard caking. The paint shall mix readily to a smooth homogeneous state.

3.4.2 Skinning. A three-quarter filled, closed 8-ounce glass jar of the paint shall show no skinning within 48 hours when tested as specified in 4.3.4.

3.4.3 Brushing properties. The paint shall have satisfactory brushing properties when tested as in 4.3.5 and shall dry to a smooth uniform film free from seeds, runs, sags, or streaks.

3.4.4 Spraying properties. The paint, tested as in 4.3.6, shall spray satisfactorily in all respects and shall show no running, sagging, or streaking. The dried film shall be smooth and uniform.

3.4.5 Dilution (Reducibility) stability. When thinned as in 4.3.7 the paint shall remain stable and uniform showing no precipitation, curdling, or separation. Slight pigment settling shall not be caused for rejection.

3.4.6 Flexibility. A film of the paint prepared and tested as in 4.3.8 shall withstand bending without cracking or flaking.

3.4.7 Knife test. A film of the paint prepared and tested as in 4.3.9 shall adhere tightly to and not flake or crack from the metal. The film shall ribbon or curl from the metal on cutting and the cut shall show beveled edges.

3.4.8 Recoating. When tested as in 4.3.10 recoating of a dried film shall produce no film irregularities.

3.4.9 Accelerated weathering. A film of paint prepared and tested as in 4.3.11 shall show no chalking, a loss of not more than 30 percent of the original 60° gloss.

3.5 Quantitative requirements.

3.5.1 The quantitative requirements of the paint shall be as specified in tables I and III.

3.5.2 The pigmentation shall be as specified in table II.

TABLE I. Quantitative requirements

Characteristics	Requirement	
	Minimum	Maximum
Coarse particle, percent by weight of pigment	----	0.5
Drying time:		
Set to touch, hours	1/2	3
Dry hard, hours	----	18
Fineness of grind	6	----
60° Gloss, specular	80	----
Water, percent by weight of paint	----	0.5

1/ Information on Rule 66 may be obtained from the Los Angeles Air Polution Control District, Los Angeles, California.

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TABLE II. Pigmentation[1/]

Color No. Fed. Std. No. 595	Color Designation	Prime Pigments
10032	Deep Brown	Red Iron Oxide, Lampblack or Carbon black
10080	Medium Brown	Yellow Iron Oxide, Red Iron Oxide, Lampblack, Titanium Dioxide.
11105	Bright Red	Toluidine Red or Quinacridone Red.
12197	Vivid Orange	Chrome Yellow, Molybdate Orange, Red Iron Oxide, Titanium Dioxide
13432	Bright Yellow	Chrome Yellow, Dinitraniline Orange, Red Iron Oxide
13538	Orange Yellow	Chrome Yellow, Dinitraniline Orange, Burnt Umber
14036	Deep Green	Chrome Green, Iron Blue, Lampblack, Titanium Dioxide
14062	Medium Green	Chrome Oxide Green, Chrome Green, Phthalo Blue
14077	Green	Titanium Dioxide, Lampblack, Chrome Yellow, Chrome Oxide Green Phthalo Blue, Zinc Oxide, Red Iron Oxide
14187	Bright Green	Chrome Yellow, Titanium Dioxide, Iron Blue
15042	Deep Blue	Iron Blue, Titanium Dioxide, Lampblack
15080	Medium Blue	Iron Blue, Titanium Dioxide, Phthalo Green, Lampblack
15102	Bright Blue	Titanium Dioxide, Phthalo Blue, Iron Blue, Alizarine Maroon
16314	Gray	Titanium Dioxide, Zinc Oxide, Lampblack, Yellow Iron Oxide
17038	Gloss Black	Carbon Black

[1/] Extender pigments may be used at the diseretion of the manufacturer.

TABLE III. Specific requirements

Color No. Fed. Std. No. 595	Total Solids Percent by Weight of paint	Pigment Solids Percent by Weight of paint	Vehicle Solids Percent by Weight of Vehicle	Consistency K U		Weight per Gallon Pounds		Contrast Ratio*
	Minimum	Minimum	Minimum	Min.	Max.	Min.	Max.	Minimum
10032	68	23	58	72	82	9.0	10.0	0.98
10080	68	23	58	72	82	9.0	10.0	0.98
11105	48	8	43	65	72	7.5	8.5	0.80
12197	55	17	45	65	72	8.5	9.5	0.93
13432	65	35	45	67	82	10.5	11.5	0.95
13538	65	35	45	67	82	10.5	11.5	0.95
14036	68	17	60	72	82	8.5	9.5	0.98
14062	68	18	60	72	82	9.0	10.0	0.98
14077	65	37	44	72	82	10.5	11.5	0.98
14187	57	31	45	67	82	9.0	10.0	0.98
15042	65	14	59	72	82	8.5	9.5	0.98
15080	68	17	60	72	82	8.5	9.5	0.98
15102	51	9	45	67	82	7.5	8.5	0.98
16473	68	38	45	72	82	10.5	11.5	0.98
17038	60	8	55	72	82	8.0	9.0	0.98

*Contrast ratio shall be determined on a draw down film of approximately 1 mil dry film thickness.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase

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order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the specification requirements specified herein unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.2 Sampling and inspection. Sampling and inspection shall be performed in accordance with method 1031 of Fed. Test Method Std. No. 141.

4.3 Test procedures.

4.3.1 Test conditions. The routine and referee testing conditions shall be in accordance with section 7 of Fed. Test Method Std. No. 141, except as otherwise specified herein.

4.3.2 Tests. The tests shall be conducted in accordance with Fed. Test Method Std. No. 141, except as otherwise indicated in Table IV.

4.3.3 Condition in container. Determine package condition on acceptance testing in accordance with method 3011 of Fed. Test Method Std. 141 and observe for compliance with 3.4.1. Reseal and agitate the can for 3 minutes on a paint shaker.[1/] On reexamination of the contents, the disclosure of any gel bodies or undispersed pigment indicated unsatisfactory settling properties.

4.3.4 Skinning. Determine skinning after 48 hours in accordance with method 3021 of Fed. Test Method Std. 141 and observe for compliance with 3.4.2.

4.3.5 Brushing properties. Determine brushing properties in accordance with method 4321 of Fed. Test Method Std. 141 for compliance with 3.4.3.

4.3.6 Spraying properties. Spray the paint as packaged or thinned with not more than 5 percent by volume of thinner conforming to TT-T-291, Grade 1, on a steel panel to a dry film thickness between 0.0008 and 0.0010 inch and observe for spraying properties in accordance with method 4331 of Fed. Test Method Std. 141. Determine compliance with 3.4.4.

4.3.7 Dilution (Reducibility) stability. (Determine the dilution stability in accordance with method 4203 of Fed. Test Method Std. No. 141 using 8 parts by volume of paint with 1 part by volume with mineral spirits conforming with TT-T-291 Grade 1. Observe for compliance with 3.4.5. Slight pigment settling shall not cause for rejection.

4.3.8 Flexibility. Determine flexibility as in method 6221 of Fed. Test Method Std. No. 141. Using a film applicator that will deposit a dry film thickness of approximately 1 mil, draw down a 2-inch wide film of paint on tin panel prepared in accordance with method 2012 of Fed. Test Method Std. No. 141. Air-dry for 18 hours then bake 2 hours at 105 ± 2°C (221° ± 4°F). Condition the panel for 1/2 hour under referee conditions. Bend over a 1/8-inch mandrel and examine for compliance with 3.4.6.

4.3.9 Knife test. Determine Knife test in accordance with method 6304 of Fed. Test Method Std. No. 141 using a flat portion of flexibility panel. Examine for compliance with 3.4.7.

4.3.10 Recoating. Apply the paint in accordance with method 4321 of Fed. Test Method Std. No. 141 on primed wood panels prepared in accordance with method 2031 using three types wood. Air-dry for 24 hours under referee testing conditions. Apply a second coat cross-wise to the first coat and air-dry as before. Examine for compliance with 3.4.8.

4.3.11 Accelerated weathering. Draw down a film of paint on duplicate flat tin panels prepared in accordance with method 2012 of Fed. Test Method Std. No. 141, with a 0.0025-inch film spplicator. Allow the coated panels to air-dry at room temperature for 96 hours. Measure the gloss and directional reflectance and subject the coated panels for 168 hours to accelerated weathering using the twin arc apparatus in accordance with method 6152 of Fed. Test Method Std. No. 141. Examine one exposed coating for chalking. Wash the exposed panel under running water with a thoroughly degreaded lamb's wool pad to remove any scum or dirt. Wipe off water

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[1/] An apparatus of this type, powered by a 1/4 hp motor, operates at a rate of 1350 shakes per minute is manufactured by Red Devil Tools, Irvington, N. J.

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with clean cheese cloth and let dry for 2 hours. Measure the gloss and reflectance of the exposed panel. Compute the percentage loss of gloss and evaluate for compliance with 3.4.9.

4.3.12 Color. Draw down a film of the paint on a white Morest or Leneta chart using .003 inch film applicator and allow to dry at room temperature for 24 hours. Determine the color in accordance with method 4250 of Fed. Test Method Std. No. 141 for compliance with 3.2.

4.3.13 Packaging, packing, and marking. The paint shall be inspected for compliance with packaging, packing, and marking requirements of section 5.

TABLE IV. Index

Characteristics	Requirement Reference	Applicable Test Method	
		Fed. Test Method Std. No. 141	Paragraph Reference
Condition in container	3.4.1	3011	4.3.3
Skinning (Partially full container)	3.4.2	3021	4.3.4
Brushing properties	3.4.3	4321	4.3.5
Spraying properties	3.4.4	4331	4.3.6
Dilution (reducibility) stability	3.4.5	4203	4.3.7
Flexibility	3.4.6	6221	4.3.8
Knife Test	3.4.7	6304	4.3.9
Recoating	3.4.8	4321	4.3.10
Accelerated weathering	3.4.9	6152	4.3.11
Color	3.2	4250	4.3.12
Consistency	Table III	4281	---
Coarse particle	Table I	4091	---
Drying time	Table I	4061	---
Fineness of grind	Table I	4411	---
60° gloss, specular	Table I	6101	---
Water content	Table I	4081	---
Weight per gallon	Table III	4184	---
Contrast ratio (Hiding power)	Table III	4122	---
Total solids, Pigment solids, and Vehicle solids	Table III	4021, 4051	---

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. The paint shall be packaged, packed, and marked in accordance with Federal Specification TT-P-143. The level of packaging shall be A, B, or C and the level of packing will be A, B, or C as specified (See 6.2). The paint shall be furnished in 1-quart metal cans, 1-gallon metal cans, 5-gallon metal pails, or 55-gallon steel drums as specified (See 6.2).

6. NOTES

6.1 Intended use. This paint is intended for general exterior use on properly primed or previously painted wood and metal surfaces.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and level of packing required (See 5.1).
(c) Unit quantity required (See 5.1).
(d) Colors as listed in Table II.

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6.3 Basis of Purchase. The paint covered by this specification should be purchased by volume, the unit being U. S. Gallon of 231 cubic inches at 60°F.

CIVIL AGENCY INTEREST

GSA
HUD

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein.

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